Journal of Molecular Liquids 339 (2021) 116762

Contents lists available at ScienceDirect

Journal of Molecular Liquids

journal homepage: www.elsevier.com/locate/molliq

Effect of the addition of deep eutectic solvent to the anthracene separation

Lan Yi [a,b], Jie Feng [a], Mario Gauthier [b], Wen-Ying Li [a,*]

[a] State Key Laboratory of Clean and Efficient Coal Utilization, Taiyuan University of Technology, Taiyuan 030024, PR China
[b] School of Chemistry and Chemical Engineering, Wuhan University of Science and Technology, Wuhan 430081, PR China

ARTICLE INFO

Article history:
Received 8 May 2021
Revised 10 June 2021
Accepted 16 June 2021
Available online 18 June 2021

Keywords:
Anthracene
Deep eutectic solvents
Separation mechanism
Intermolecular interaction

ABSTRACT

Anthracene (AN) is an important fine chemical feedstock, mainly derived from the anthracene oil fraction of high-temperature coal tar. The traditional solvent crystallization method used to obtain anthracene involves large amounts of solvents. A method with a high separation capacity and a high selectivity needs to be developed urgently. To this end, the composition of crude AN oil and the solubility of its main components in xylene were determined, and phenanthrene in crude AN oil was extracted with xylene. Pure deep eutectic solvents (DESs) and DESs–organic solvent mixture were then applied to separate anthracene and carbazole, respectively. Based on the distribution coefficient, selectivity, and purity of the AN product, the influence of organic solvent type, the molar ratio of DESs to organic solvent, and DESs type on the separation efficiency were investigated. Finally, the separation mechanism was investigated with density functional theory calculations and UV–visible spectroscopy. The results show that the phenanthrene content decreased from 25.5% to 2.6% and most of other condensed aromatics in crude AN were removed after washing twice with xylene. DESs–organic solvent mixtures had better performance than pure DESs for the separation of anthracene and carbazole, and under optimal condition, the 97.5% purity of AN is acquired in a single separation step with a tetraethyl ammonium chloride–glycerol DESs–DMF mixture. The more complicated interactions between phenanthrene and xylene than between anthracene / carbazole and xylene favored the removal of phenanthrene. The hydrogen bond with the minimum bond energy (-17.6 kJ/mol) between DESs and carbazole is also stronger than that with the maximum bond energy (-16.8 kJ/mol) between DESs and anthracene. Adding DESs to organic solvent can enhance the interactions between carbazole and the solvent, but had no significantly influence on interactions between anthracene and the solvent, thereby facilitating the separation of anthracene and carbazole.
© 2021 Elsevier B.V. All rights reserved.

1. Introduction

Crude anthracene oil, a fraction distilled from high-temperature coal tar, is a mixture consisting of condensed aromatic compounds with 2–5 aromatic rings [1]. Among the condensed aromatics are anthracene (AN), carbazole (CAR), and phenanthrene (PHE), which account for more than 1 wt% of high-temperature coal tar. AN, one of the most valuable components in coal tar, is an irreplaceable raw material in the manufacture of dyes, bactericides, pesticides, light-emitting diodes, and so on [2,3]. For the production of fine chemicals, AN with a purity higher than 98% is essential. The remaining 2% impurities, which are difficult to remove, include CAR [4]. Much effort is need to improve the purity of AN while refining crude AN oil, as the separation and purification of AN is usually accompanied by the simultaneous removal of CAR.

Commonly used methods to isolate AN include distillation, supercritical fluid extraction, zone melting, emulsion membrane separation, and solvent crystallization [5–7] and each method has its own advantages and disadvantages [7,8]. For example, distillation has the advantage of continuous operation and the disadvantage of high energy consumption; supercritical fluids extraction has a high extraction efficiency for aromatics, but require harsh operating conditions; the zone melting technique is complex to operate, and the emulsion membrane separation method is unstable. Ionic liquids (ILs), known as green solvents, are also being investigated in the refining of crude AN oil, and it was shown that 66.6% recovery and 90.2% purity of CAR separated from anthracene oil could be achieved [9]. However, as a new separation technology, the use of ILs in crude AN oil refining still faces difficulties in implementation [10], requiring further investigations and studies.

* Corresponding author.
E-mail address: ying@tyut.edu.cn (W.-Y. Li).

https://doi.org/10.1016/j.molliq.2021.116762
0167-7322/© 2021 Elsevier B.V. All rights reserved.

Nomenclature

Abbreviation

AIM	atoms in molecules
AN	anthracene
AU	atom unit
BCPs	bond critical points
BSSE	basis set superposition error
CAR	carbazole
ChCl	choline chloride
D	Debye unit
Da	atomic mass unit, Dalton
DESs	deep eutectic solvents
DFT	density functional theory
DMA	N,N-dimethylacetamide
DMF	N,N-dimethylformamide
E_{bond}	energy of hydrogen bond
ESP	electrostatic potential
GC	gas chromatography
IGM	independent gradient model
LOL	localized orbital locator
LOL-π	LOL purely contributed by π-orbital
LOLIPOP	LOL integrated π over plane
NMP	N-methylpyrrolidone
PHE	phenanthrene
RDG	reduced density gradient
$sign(\lambda_2)\rho(r)$	the product of the sign (λ_2) function, which is the sign of the second largest eigenvalue λ_2 of the electron density Hessian matrix, multiplied by the value of the electron density
SI	Supporting Information
SS	solubility selectivity
TEAC	tetraethylammonium chloride
TPAC	tetrapropylammonium chloride
VDW	van der Waals

Greek alphabet

ρ	the electron density
$\nabla^2\rho$	Laplacian of the electron density

Solvent crystallization is a common but efficient separation and purification method because it is a low energy consumption and easy process. Unfortunately, repeated crystallization processes consume large amounts of solvent. Due to the similarity in structure of AN and CAR, solvent extraction processes cannot guarantee large distribution coefficients and high selectivity. In recent studies, the AN purity attained by solvent extraction typically reached 90–95% [11], with a maximum purity of about 97% [12]. Due to the limitations of these separation techniques, a new method using a third component in traditional organic solvents was proposed and yielded promising results. For instance, adding supercritical CO_2 to acetone promoted the separation of AN and PHE [13]; dissolving an amide/amine combination in N,N-dimethylformamide (DMF) improved the solubility of PHE in that solvent [11]. In research by Ye et al. [7,14], it was shown that the addition of isopropanolamine, urea, and H_2O to DMF improved the separation selectivity for AN and CAR significantly. The enhanced separation efficiency was ascribed to the electron-rich atoms or groups in the additives, introducing intermolecular interactions such as hydrogen bonds with the –NH group of CAR [14]. Deep eutectic solvents (DESs), similar to ILs, have a broad liquid temperature range, a low vapor pressure, a high chemical stability, and were used for the removal of CAR from fuels [15]. Due to the formation of hydrogen bond between the DESs and CAR, efficient fuel denitrogenation was achieved. DESs include both hydrogen bond donors and hydrogen bond acceptors, facilitating the formation of intermolecular interactions with target components. Correspondingly, the performance of DESs was excellent in the extraction and isolation of many components including phenolic compounds [16,17], nitrogen/sulfur- containing compounds [18], benzene derivatives [19], and metals [20]. The application of DESs as solvents or additives to refine crude AN oil has not been reported, however.

In this study, the solubility of AN, PHE, and CAR in xylene were determined and PHE was first extracted from crude AN oil with xylene. DESs were then used as separation agents or additives in organic solvent for the separation of AN and CAR. The separation efficiency of DESs, either pure or mixed with solvents was investigated, as well as the separation mechanism involved.

2. Experimental and calculation section

2.1. Chemical reagents

Analytical pure Anthracene (AN), carbazole (CAR), and phenanthrene (PHE) were purchased from Macleans Reagent Co., Ltd. Glycerol (99 wt%), levulinic acid (99 wt%), malonic acid (99.5 wt %), 9-fluorenone (98 wt%), Tetraethyl ammonium chloride (TEAC, 98 wt%), and tetrapropylammonium chloride (TPAC, 98 wt%) were purchased from Shanghai Aladdin Biochemical Technology Co. Analytical pure N,N-dimethyl formamide (DMF), N-methyl pyrrolidone (NMP), N,N-dimethylacetamide (DMA), hexane, and xylene were purchased from Tianjin Kemiou Chemical Co. Acetonitrile (HPLC pure) was purchased from Shanghai Aladdin Biochemical Technology Co.

Crude anthracene oil was provided by one steel factory, Wuhan, China. The composition was detected by GC–MS (7890B GC system, 5977A MSD; Agilent Technology).

2.2. Preparation of DESs and AN-CAR mixture

The preparation procedure for TEAC–glycerol DESs was provided herein as an example, where TEAC and glycerol were selected as the HBA and HBD, respectively. In a typical process, TEAC and glycerol were mixed at equimolar ratio in a capped conical flask and agitated with magnetic stirring at 80 °C until the liquid became homogeneous. Other DESs were prepared in a similar way.

AN–CAR mixture was prepared by dissolving AN and CAR in hot benzene and evaporating benzene after AN and CAR were completely dissolved.

2.3. Separation experiments

Separation of AN and CAR. The AN–CAR mixture was added to the prepared DESs and stirred magnetically for 30 min at the set temperature. Filtered them to obtain cake and filtrate at room temperature. A small amount of cake and filtrate were respectively collected and analyzed by the UV–visible spectrophotometer. The

distribution coefficient and selectivity for CAR vs. AN were calculated by equations (1) and (2):

$$\text{Distribution coefficient } (D_i) = \frac{C_i^T}{C_i^C} \quad (1)$$

$$\text{Selectivity } \left(S_{\frac{CAR}{AN}}\right) = \frac{D_{CAR}}{D_{AN}} \quad (2)$$

where c_i^C and c_i^T are the contents of AN or CAR in filter cake and filtrate, respectively. D_i is the distribution coefficient of CAR or AN, and i is AN or CAR. $S_{CAR/AN}$ is the selectivity for CAR vs. AN.

Separation of crude anthracene oil. The crude anthracene oil mainly contains 34.1 wt% AN, 25.5 wt% PHE, and 16.4 wt% CAR. The separation experiment on crude anthracene oil was conducted as follows: 100 g crude anthracene oil and 60 ml xylene were mixed in a 500 ml flask equipped with a reflux condenser pipe and stirred magnetically for 30 min at 120 °C in an oil bath. After the crude anthracene oil was completely dissolved in xylene, this mixture was filtered at 30 °C to obtain the first filter cake. Subsequently, the first filter cake was mixed with xylene at the liquid/-solid ratio of 1.5:1 and the conditions of stirring and filtering were the same as the above. And the second filter cake rich in AN and CAR was obtained and dried at 40 °C for 12 h. The dried second filter cake was mixed with DESs–DMF solvent, magnetically stirred at a set temperature for 30 min, and filtered at room temperature to get AN product. Take a small amount of product for GC detection to determine the purity.

2.4. Analytical methods

The content of AN and CAR in cake and filtrate for measuring distribution coefficient was analyzed by UV–visible spectrophotometer (CARY 300). The UV–vis absorption peaks of CAR and AN at 292 nm and 378 nm had no overlap and no mutual interference (**Figure S1** of supporting information) and they were selected as the characteristic absorption wavelengths of CAR and AN respectively for quantitative calculation. UV–visible spectral experiments were also carried out to verify the interactions between solvent and AN or CAR. The sample scan range was 220 to 420 nm.

The compositions of crude AN oil and AN product were detected by GC–MS (7890B GC system, 5977A MSD; Agilent Technology), equipped with a HP-5 capillary column (30 m × 0.32 mm × 0.25 μm) and a flame ionization detector (FID). The chromatographic detection temperature conditions were 280 °C for the over and injector, and 300 °C for the detector. The MS detection conditions were a temperature of 250 °C for the detector, 70 eV emission voltage, and mass range of 33–400 Da.

2.5. DFT calculations

DFT calculations were performed using G09 program package [21]. More than five initial geometries for compound or conformation were optimized under no symmetry restrictions at the M06-2X/6–31 + G(d,p) level. The frequency was analyzed at the M06-2X/6–31 + G(d,p) level and lack of imaginary frequencies for the optimized structures indicated minimum energy structures. The single-point energy was computed at the M06-2X/6–311++G (d, p) level and the counterpoise procedure [22] was considered to correct the basis set superposition error (BSSE). The interaction energy, ΔE, was computed using equation (3):

$$\Delta E = E(A\cdots B) - E(A) - E(B) + BSSE \quad (3)$$

Localized orbital locator (LOL) [23] integrated π over plane (LOLIPOP) and LOL-π map were used to measure and analyze π-stacking ability and aromaticity of AN, PHE and CAR. Electrostatic potential (ESP) [24] was used to analyze nucleophilic and electrophilic sites of AN, PHE and CAR. Reduced density gradient (RDG) [25] was used to analyze the weak interaction between xylene and AN or PHE or CAR. Independent gradient model (IGM) [26] was used to visually study the interactions between DESs and AN, CAR, or DMF. Atoms in molecules (AIM) population [27] was used to identify the hydrogen bond and van der Waals (vdW) interaction. The wave function was analyzed using the program Multiwfn 3.6 [28] and all the isosurfaces were described using VMD 1.9.2 [29].

3. Results and discussion

3.1. Removal of PHE from crude AN oil

The composition of crude AN oil is shown in Fig. 1. PHE (25.5%), AN (34.1%), and CAR (16.4%) were the main components of crude AN oil, with small amounts of fluorene, dibenzofuran, dibenzothiophene and other fused ring aromatic compounds. Note that 9-fluorenone was used as internal standard for the analysis. Xylene, an aromatic solvent, was used to extract PHE from the crude AN oil. Before the separation, the solubility of AN, PHE, and CAR in xylene and the solubility selectivity for PHE vs. CAR (AN) were determined as described in the Supporting Information (SI); the result obtained summarized in Fig. 2. It can be seen that at 30 °C, the solubility of PHE in xylene was nearly 38 times higher than CAR and 18 times higher than AN, indicating that xylene is capable of removing PHE from crude AN oil. Similarly to PHE, xylene is a weakly polar molecule, so it can dissolve a certain amount of PHE in agreement with the "like dissolves like" rule. The dipole moments calculated (at m062x/def2-TZVPD level) for AN, CAR, PHE and xylene were 0, 1.66, 0.01, and 0.42 D, respectively. AN is a non-polar molecule due to its completely symmetrical structure, while CAR is strongly polar due to its –NH bond. Clearly, the interactions between xylene and AN or CAR are insufficient to compete with the interactions between AN molecules or CAR molecules, resulting in low solubility of AN and CAR in xylene. At increased temperatures, the solubility of PHE, AN, and CAR in xylene increased to different extents, but the solubility selectivity (SS) for PHE vs. CAR (AN)decreased. To ensure the efficient removal of PHE, the crystal filtration temperature was therefore set to 30 °C. The PHE was removed best by washing twice with xylene, because for a constant total amount of solvent, the operation efficiency is higher according to the "small quantity, many times" rule [30]. The GC analysis results for crude AN oil after the removal of PHE are shown in **Figure S2** of SI. After removing PHE with xylene, the AN content in crude AN oil increased from 34.1% to 68.9%, the CAR content increased from 16.4% to 22.2%, and the PHE

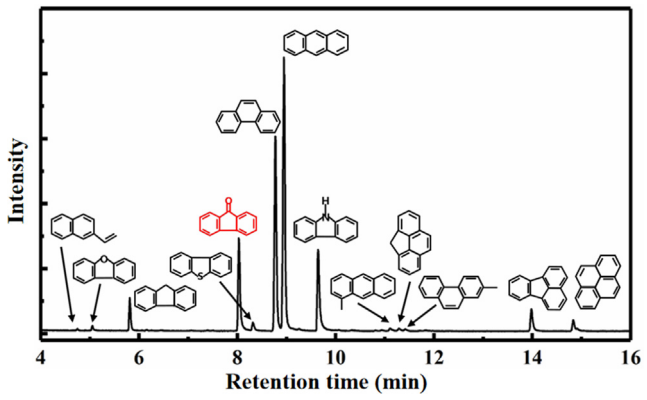


Fig. 1. Composition analysis of crude anthracene oil.

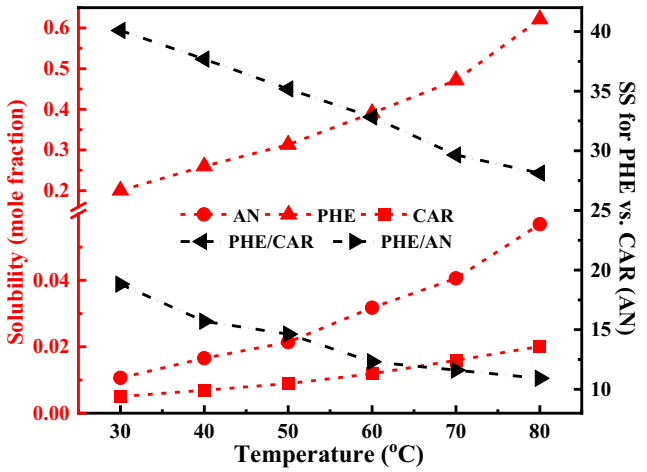


Fig. 2. Temperature-dependent solubility of AN, PHE and CAR, and solubility selectivity for PHE vs. AN (CAR) in xylene.

content decreased from 25.5% to 2.6%. As compared with the crude AN oil (Fig. 1), most of PHE, fluorene, dibenzothiphene, dibenzofuran and other impurities in crude AN oil were removed with xylene.

3.2. Separation of anthracene and carbazole

3.2.1. DESs for the separation of AN and CAR

After the removal of PHE with xylene, the main remaining components of AN oil were AN and CAR. The DESs, as a new type of green solvents, have been widely used in the chemical industry and with good performance [17,31,32]. Ammonium-based DESs are excellent solvents to separate both basic and non-basic nitrogen compounds from fuels [15], and glycerol is a natural material and market saturated. Therefore, tetraethylammonium chloride and glycerol were used as hydrogen bond acceptor and hydrogen bond donor, respectively, to prepare DESs for the separation of AN and CAR; the separation performance of that system is summarized in Fig. 3. Clearly, the distribution coefficient and selectivity were strongly dependent on the separation temperature (Fig. 3a). For increasing temperatures, the distribution coefficient of CAR increased from 0.083 at 30 °C to 0.187 at 80 °C, while that of AN only increased slightly as compared with CAR. The viscosity and surface tension of the DESs gradually decreased with the temperature [33], making it easier for AN and CAR to partition into DESs at high temperatures, thereby increasing the distribution coefficient. A maximum in the selectivity curve (69.0) was reached at 60 °C, due to the different rates of increase in the distribution coefficients of AN and CAR. As the DESs dose was increased, the distribution coefficient of CAR decreased within the range investigated, but a peak also appeared in the selectivity curve (Fig. 3b). The concentration of CAR in the DESs decreased for increased DESs doses, which would reduce the distribution coefficient of CAR. In general, a large distribution coefficient and high selectivity favor the DESs-promoted separation of AN and CAR. Unfortunately, the high viscosity of the DESs could cause problems such as a low separation capacity, low operation efficiency, and equipment blockage when used directly for the separation of AN and CAR.

3.2.2. Mixture of DESs and organic solvent for the separation of AN and CAR

Due to their high separation capacity, traditional solvents such as DMF was commonly used for the separation AN and CAR, but their low selectivity requires repeated crystallization cycles and leads to the consumption of large amounts of solvent [12]. To combine the advantages of the large separation capacity of traditional solvents with the high selectivity of DESs, these two components were mixed for the separation of AN and CAR.

Effect of organic solvent type. TEAC–glycerol DESs were selected in combination with three common organic solvents (DMF, DMA, and NMP) to investigate the separation of AN and CAR, as presented in Fig. 4, where the ordinate represents the molar amount of DESs in the mixed solvent, and the abscissa represents the distribution coefficient of CAR (Fig. 4a) and the selectivity for CAR vs. AN (Fig. 4b). Interestingly, both the distribution coefficient and the selectivity were higher for DESs-organic solvent mixtures than for pure DESs (Fig. 3). This may be due to that, as compared with pure DESs, the viscosity of the mixed system was significantly reduced, facilitating the phase transfer of CAR. When adding 50% (molar amount) of DESs, the distribution coefficient of CAR in the three mixed solvents were not very different, ranging from 0.492 to 0.531 (Fig. 4a); when the amount of DESs was reduced to 10%, the distribution coefficient of CAR in the three mixed solvents increased to 0.927 ~ 1.170, reaching a maximum value for the DESs–DMF mixture. In solvent mixtures with different amounts of DESs, the selectivity for CAR vs. AN followed the order DESs–DMF > DESs–DMA > DESs–NMP (Fig. 4b), with a maximum selectivity for CAR vs. AN (10% DESs–DMF, 231.2) more than 5 times the minimum selectivity (10% DESs–NMP, 40.0). More-

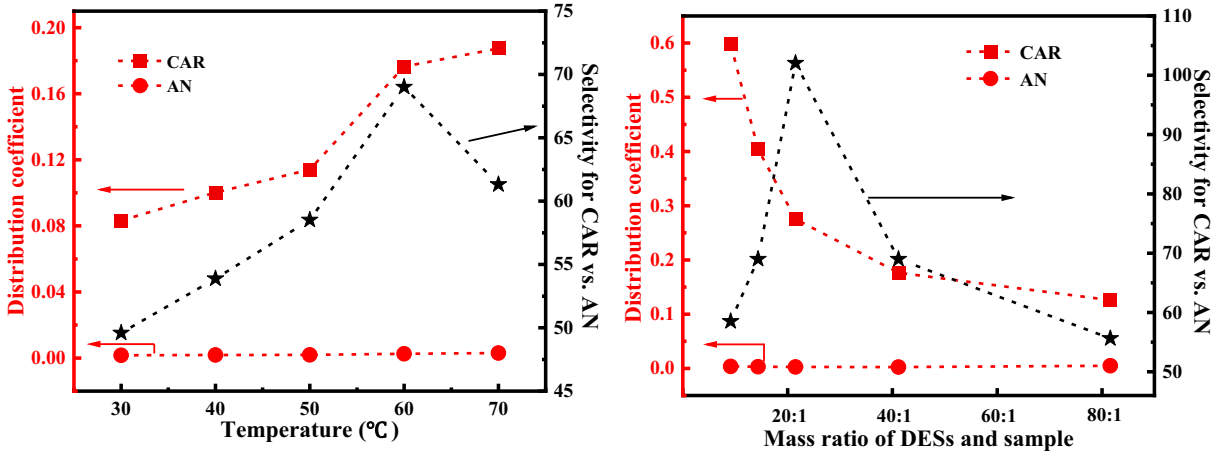


Fig. 3. Effects of temperature (a) and TEAC-glycerol DESs dose (b) on the separation performance.

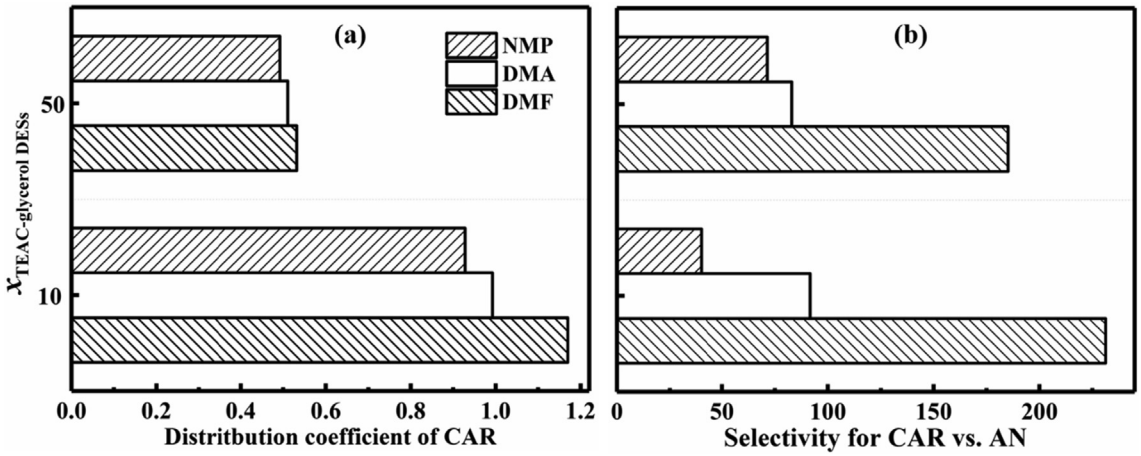


Fig. 4. Effect of organic solvent type on separation with DESs–solvent mixture.

over, for the maximum selectivity for CAR vs. AN, the result of DESs–DMF (231.2) was higher than the results of pure DESs (104.8, Fig. 3b) and pure DMF (16.9) [7]. From the perspective of distribution coefficient and selectivity, the DESs–DMF system had the best separation performance.

Effect of DESs–DMF molar ratio. Since the molar ratio of DESs and DMF in the mixture had a certain influence on the separation performance (as shown in Fig. 4), the influence of the DESs and DMF molar ratio on the separation of AN and CAR was investigated in more detail, and the results obtained are shown in Fig. 5, where the ordinate represents the distribution coefficient of CAR or the selectivity for CAR vs. AN, and the abscissa represents the molar amount of DESs in DESs–DMF mixture. For increased molar amounts of DESs in the solvent mixture, the distribution coefficient of CAR increased from 1.20 (1% DESs) to 1.29 (5% DESs), and then decreased to 0.38 (80% DESs). Similarity, the selectivity for CAR vs. AN also increased from 89.9 to 231.2, and then decreased to 122.5 with the molar amount of DESs increased. There was a maximum in both the distribution coefficient and selectivity curves. This trend is consistent with results reported for the separation of tocopherol isomers using ionic liquid–organic solvent mixtures [34]. According to Yang et al. [34], the dipolarity and hydrogen bond basicity, two Kamlet-Taft empirical parameters characterizing the interactions between a solvent and a solute [35–37], would change for the DESs–DMF mixed solvent system as compared with pure DMF. As the molar ratio of DESs in DMF increase, the polarity of the mixture increased gradually, but its hydrogen bond basicity increases sharply to a maximum for a low amount of added DESs, and it remains constant thereafter [34,38]. The dipolarity and hydrogen bond basicity have a synergistic effect on the separation of substances [12], which makes the inflection points appear in the distribution coefficient and selectivity curves. It is obvious that adding a small amount of DESs to DMF suffices to achieve a high distribution coefficient and selectivity. Consequently, the molar amount of DESs used in the mixed solvent system was set to 10%.

Effect of DESs type. Taking advantage of flexibility in the characteristics of DESs, a total of four DESs were designed with TEAC and tetrapropylammonium chloride (TPAC) as hydrogen bond acceptors, and glycerol, levulinic acid and malonic acid as hydrogen bond donors, with the 1:1 molar ratio of hydrogen bond acceptor and hydrogen bond donor. Noted that some DESs such as choline chloride–glycerol and DMF are immiscible, the choice of hydrogen bond acceptors and hydrogen bond donors is not endless herein. The DESs were mixed with DMF for the separation of AN and CAR, and the results obtained are summarized in Fig. 6. The purity of the isolated AN component

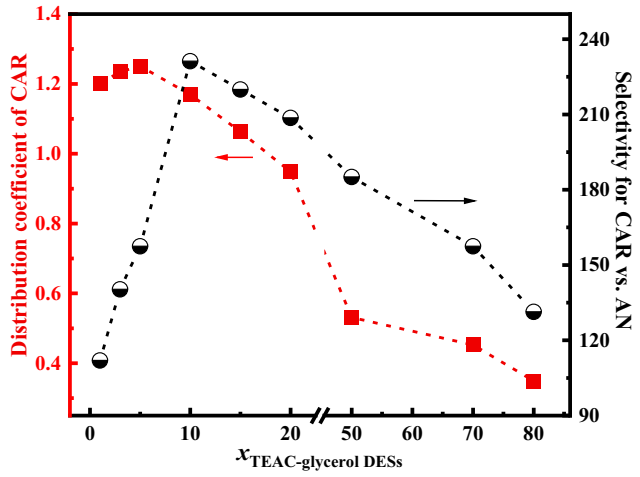


Fig. 5. Effect of the DESs–DMF molar ratio on the separation process.

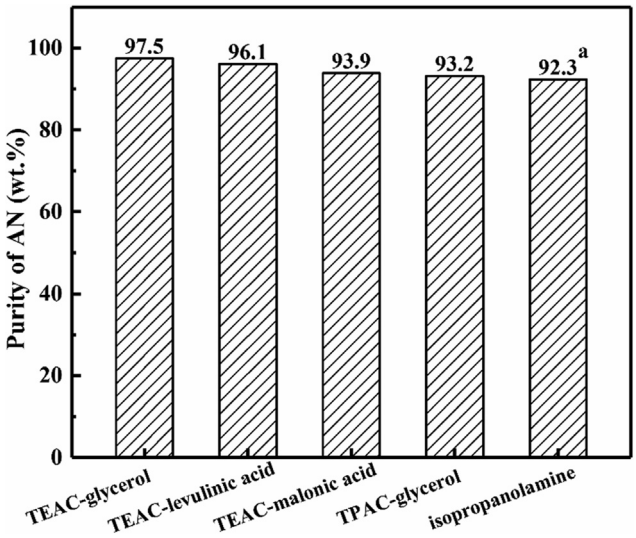


Fig. 6. Influence of DESs type on AN purity. [a] Result from Ref. [7]

followed the order TEAC–glycerol DESs–DMF (97.5%) > TEAC–levulinic acid DESs–DMF (96.1%) > TEAC–malonic acid DESs–DMF (93.9%) > TPAC–glycerol DESs–DMF (93.2%). As compared with the purity of AN (92.3%) obtained using a isopropanolamine–DMF mixed solvent system [7], the mixture of DMF with TEAC–malonic acid DESs or TPAC–glycerol DESs seem to have no obvious advantage for the separation of AN and CAR, since the purity of AN was only increased by 1.6% and 0.9%, respectively. However, due to the formation of a solid AN–CAR mixture [39,40], their completely separation in a single separation step represents a huge challenge [41]. To achieve a high (greater than 97%) AN purity, it is usually necessary to use 5 ~ 8 separation steps in the solvent-based procedure [7,12], while an anthracene purity reaching 97.5% was achieved in a single step with the TEAC–glycerol DESs–DMF system. This AN purity (97.5%) was also higher than the result (93.7%) obtained by using pure DMF separation. Therefore, it can be stated that DESs are appropriate components in mixed solvent system with DMF for the efficient separation of AN and CAR. The GC analysis result for the AN component is shown in **Figure S3** of SI. The AN component is obviously dominant, with only a small amount of CAR present.

3.3. Separation mechanism

3.3.1. Structural characteristics of AN, PHE and CAR

PHE and AN are isoelectronic conjugate systems, and CAR is a multi-electron conjugate system; all these molecules have seven π-orbitals. Their π-electron characteristics were investigated using of localized orbital indicator (LOL) analysis [23], presented in Fig. 7. The LOL-π (LOL purely contributed by π-orbitals) of the three six-membered rings of AN are connected, implying that delocalization of the π-electrons occurs among the three rings, and that the aromaticity of the three rings is similar. The LOL integrated π over plane (LOLIPOP) value indicating the π-stacking strength of the aromatic rings [42], are 6.22 and 5.92 for the outer rings and the central ring of AN, respectively, which shows that the tendency for π-stacking is not very different over the three rings. On the contrary, the LOL-π of the three six-membered rings of PHE is not connected, and there are three disconnections in the central ring (shown by the black arrows in Fig. 7b). These results show that extent of π-electron conjugation (aromaticity) is higher for the outer rings than the central ring. The LOLIPOP values are 7.21 and 5.74 for the outer and central rings of PHE, respectively, meaning that the π-stacking tendency for the central ring is stronger than for the outer rings. The LOL-π of the three six-membered rings in CAR is similarly disconnected in three locations on the central five-membered ring (shown by the black arrows in Fig. 7c), which demonstrates that π-electron delocalization in CAR mainly occurs in the two outer rings. The LOLIPOP values are 7.20 and 3.70 for the outer rings and the central ring of CAR, respectively, showing again that the π-stacking tendency was higher for the central ring than for the outer rings. The π-stacking ability of the central and outer rings of CAR is significantly larger than for PHE.

The electrostatic potential (ESP), a useful tool to intuitively understand the type and intensity of electrostatic interactions between molecules [43], was analyzed for AN, PHE and CAR and the results obtained are shown in **Figure S4** of SI. The order of ESP distribution ranges was CAR (-83.1 ~ 202.0 kJ/mol) > PHE (-68.7 to 92.1 kJ/mol) > AN (-66.5 to 71.5 kJ/mol). The broad ESP distribution of CAR indicates that it is more likely to interact with very polar solvents as compared with AN and PHE. The global ESP minima all appeared near the outer rings (**Figure S4** of SI) for all three molecules, which contributes to the formation of π-π stacking interactions among them and with solvent molecules.

3.3.2. Removal of PHE

The reduced density gradient (RDG) [25] is a very popular method to study weak interactions in real space, based on the electron density and its derivatives. Color-mapped scatter diagrams of RDG versus sign $(\lambda_2)\rho$ for complexes of xylene–AN, xylene–PHE, and xylene–CAR are compared in Fig. 8. These results can be used to depict noncovalent interactions between xylene and AN (PHE, CAR) with colored scatter diagrams where locating RDG = 0.5 a. u., blue spike represents hydrogen bond, brow and green spike imply van der Waals (vdW) interactions, and red spike indicates steric repulsions [25]. The interactions between xylene and the three solutes are different: For the xylene–PHE complex, four brown-green or green spikes ranging from −0.02 to 0.02 suggesting vdW interactions are observed, and the red spike around 0.02

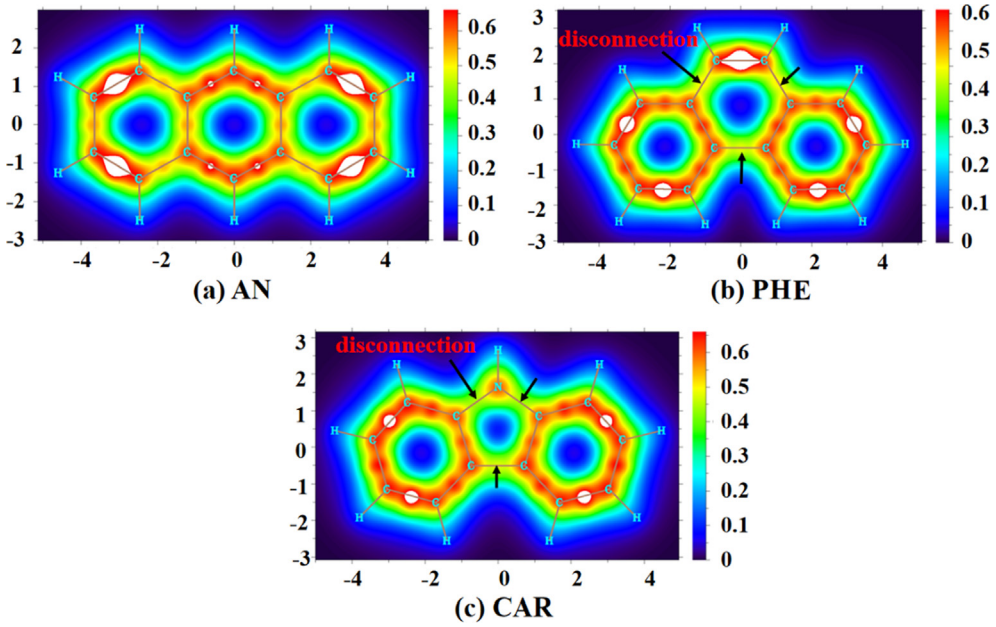


Fig. 7. LOL-π maps for AN, PEH, and CAR (length unit: Å).

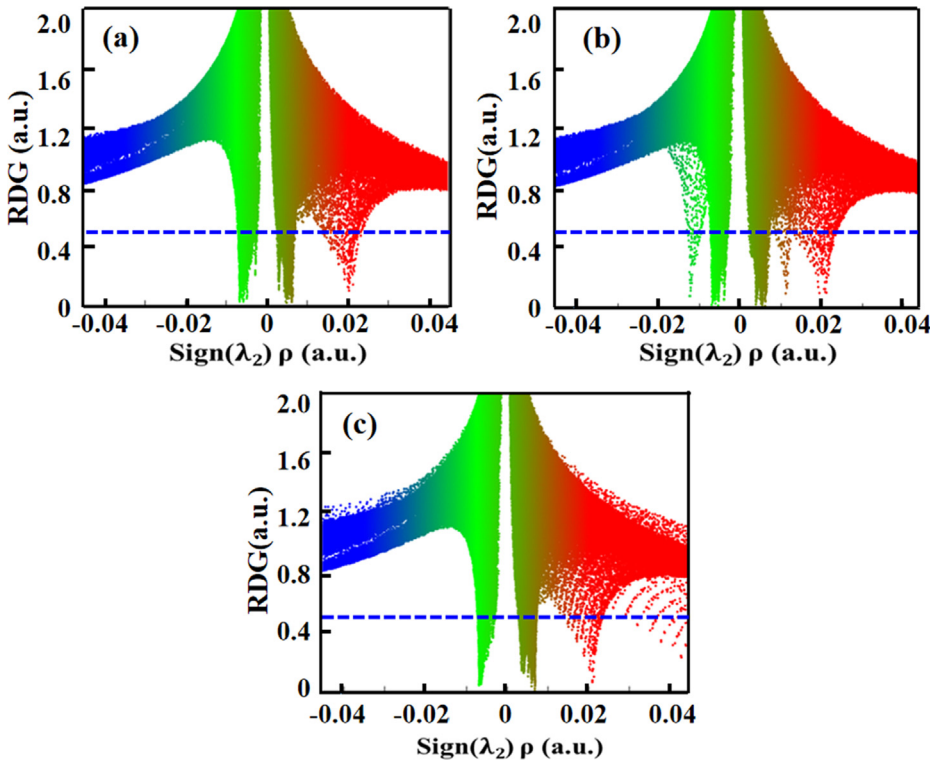


Fig. 8. Color mapped RDG scatter diagrams for complexes of xylene with AN (a), PHE (b), and CAR (c). RDG (a.u.) represents the RDG function value, and sign (λ_2)ρ (a.u.) represents the product of the sign of the eigenvalue λ_2 of the electron density Hessian matrix and the value of the electron density function.

shows that steric repulsion is also present. In contrast, for complexes of xylene–AN and xylene–CAR, besides steric repulsion (spike at 0.02), only two vdW interactions (spikes from −0.01 to 0.01) are dominant. These results demonstrate that the interactions between xylene and PHE are more complex than between xylene and AN or CAR, making the solubility of PHE in xylene highest, which is conducive to the removal of PHE from crude AN oil with xylene.

3.3.3. Separation of AN and CAR

The –NH group in the CAR molecule easily forms hydrogen bonds with strongly electronegative atoms or groups. As indicated by IGM analysis [26] (Fig. 9), the –NH of CAR and Cl^- of the DESs can form a N–H…Cl hydrogen bond, and the benzene rings of CAR form C–H…π interactions with the choline cation of DESs, but C–H…O and C–H…π appear to dominate the interactions between AN and the DESs. AIM analysis [44] was used to study the critical points between DESs and AN or CAR, and to quantify the bond interactions in terms of the electron density (ρ(r)), the Laplacian of the electron density ($\nabla^2\rho(r)$), and the bond energy (E_{bond}) at the bond critical point (BCP) [45]. The BCP is a measure of bond interactions, and the values of ρ(r) and $\nabla^2\rho(r)$ describe the strength and the characteristics of a bond, respectively. The hydrogen bond critical ranges are 0.002–0.035 a.u. for ρ(r) and 0.024–0.139 a.u. for $\nabla^2\rho(r)$, the vdW interaction critical ranges are 0–0.002 a.u. for ρ(r) and 0–0.024 a.u. for $\nabla^2\rho(r)$. The values of ρ(r), $\nabla^2\rho(r)$, and E_{bond} calculated for BCPs between DESs and AN or CAR are provided in Table 1. For the DESs–CAR complex, five BCPs were identified and the values of $\nabla^2\rho(r)$ were distributed

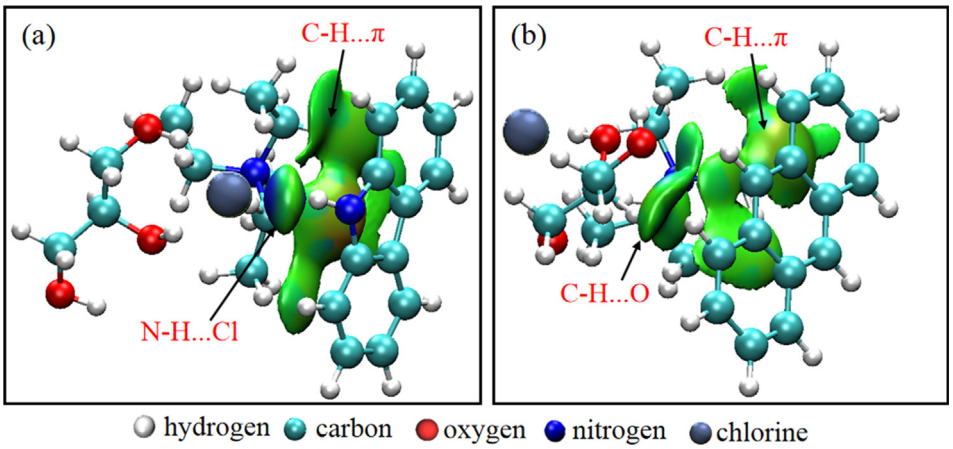


Fig. 9. IGM analysis of TEAC–glycerol DESs and CAR (a), or AN (b).

Table 1
Topological analysis of carbazole–DESs and anthracene–DESs complexes.

bond	ρ(r)/a.u.	$\nabla^2\rho(r)$/a.u.	E_{bond} (kJ/mol)
TEAC–glycerol DESs–CAR			
$H_{66}...Cl_{31}$	0.0217	0.0668	−36.5
$H_{16}...C_{49}$	0.0116	0.0396	−17.6
$H_{19}...C_{56}$	0.0072	0.0211	−10.2
$H_{34}...C_{47}$	0.0073	0.0196	−8.9
$H_{29}...C_{59}$	0.0054	0.0147	−6.6
TEAC–glycerol DESs–AN			
$H_{56}...O_{13}$	0.0103	0.0341	−16.8
$H_{22}...C_{46}$	0.0090	0.0270	−12.6
$H_{60}...O_{13}$	0.0081	0.0240	−12.3
$H_{39}...C_{52}$	0.0074	0.0208	−9.5
$H_{6}...C_{55}$	0.0064	0.0190	−8.7
$H_{26}...C_{51}$	0.0068	0.0179	−8.4
$H_{35}...C_{46}$	0.0060	0.0153	−7.1
$H_{32}...C_{50}$	0.0046	0.0126	−5.5

from 0.0147 to 0.0668 a.u., showing that the two BCPs were hydrogen bonds, while the other three were vdW interactions. The E_{bond} calculated for the two hydrogen bonds were −36.5 and −17.6 kJ/mol, respectively. For the DESs–AN complex there were eight BCPs, with values of $\nabla^2\rho(r)$ (0.0126 to 0.0341 a.u.) indicating that three hydrogen bonds were formed between DESs and AN with E_{bond} values of −12.3, −12.6, and −16.8 kJ/mol. Although there were more hydrogen bonds between the DESs and AN, their maximum E_{bond} (-16.8 kJ/mol) was lower than the minimum E_{bond} between DESs and CAR (-17.6 kJ/mol), which provided the possibility of using DESs to separate AN and CAR. The interactions of TEAC-glycerol DESs with DMF were also discussed using IGM analysis [26], as shown in **Figure S5** of SI. DMF mainly reacts with choline ion and chloride ion of DESs by interactions of C–H...O and C–H...Cl.

The UV–visible spectra obtained for CAR and AN in n-hexane, DMF, TEAC–glycerol DESs, and DESs–DMF mixture are compared in Fig. 10. These results can serve to probe the interaction strength between the solvent and AN or CAR molecules through shifts in the UV absorption peaks: The greater the shift, the stronger the interactions between the solvent and AN or CAR molecules. For CAR (Fig. 10**a**), three UV peaks appeared at 330, 318, and 290 nm in *n*-hexane, while they appeared at 337, 324, and 293 nm in DMF, 339, 326, and 294 nm in DESs, and 339, 325, and 294 nm in DESs–DMF mixed solvent system. Relatively to the UV peaks of CAR in *n*-hexane, the peaks shifted by 4 ~ 9 nm, 3 ~ 7 nm, and 4 ~ 9 nm in DESs, DMF, and DESs–DMF, respectively. Based on this, the magnitude of the interactions between CAR and the three solvents varied in the order DESs ≈ DESs-DMF mixture > DMF, showing that the interactions between CAR and the mixed solvent system was comparable to CAR and DESs. For AN (Fig. 10**b**), in comparison with the five UV peaks (375, 355, 338, 323, and 308 nm) in *n*-hexane, the peaks shifted by 5 ~ 6 nm, 3 ~ 4 nm, and 3 ~ 4 nm in DESs, DMF, and DESs–DMF, respectively. The results indicate that the interactions between AN and the three solvents varied in the order DESs > DESs-DMF mixture ≈ DMF, and therefore that the interactions between AN and the mixed solvent system were close to those between AN and DMF. The DFT calculations showed that the interaction energies between the DESs and CAR (-88.3 kJ/mol) or AN (-58.0 kJ/mol) were larger than between DMF and CAR (-49.0 kJ/mol) or AN (-35.5 kJ/mol). Combining the results of the UV–visible analysis and the DFT calculations, the interactions between the DESs–DMF mixture and CAR were strength than between the DESs–DMF mixture and AN. Therefore, adding DESs to DMF enhanced the interactions between CAR and the solvent, but had no evident effect on the interactions between AN and the solvent. In other words, the DESs widened the gap in interaction strength between the separation solvent and CAR or AN, thereby promoting the separation of AN and CAR.

4. Conclusions

We presented a method to separate anthracene from crude anthracene oil using mixed solvent system of DESs and organic solvents. This approach combines the advantages of the high selectivity of DESs and the large separation capacity of organic solvents. Xylene can form intermolecular interactions with phenanthrene easily, and the interactions are more complex than between xylene and anthracene or carbazole, which facilitates the removal of phenanthrene. When separating anthracene and carbazole, pure DESs achieved a high distribution coefficient and selectivity, but the separation performance was not as good as for the DESs-organic solvent mixture. Among the different conditions investigated, optimal separation performance was achieved at a molar ratio of DMF to TEAC–glycerol DESs of 9. These conditions yielded anthracene with a purity level of 97.5% in a single step, higher than achieved with an isopropanolamine–DMF mixed solvent system (92.3%). Due to the isolation of a solid product, the separation efficiency with the DESs–DMF mixed solvent is considered to be excellent. The efficient separation achieved with the DESs–DMF mixed solvent system is ascribed to N-H...Cl and C-H...π interactions

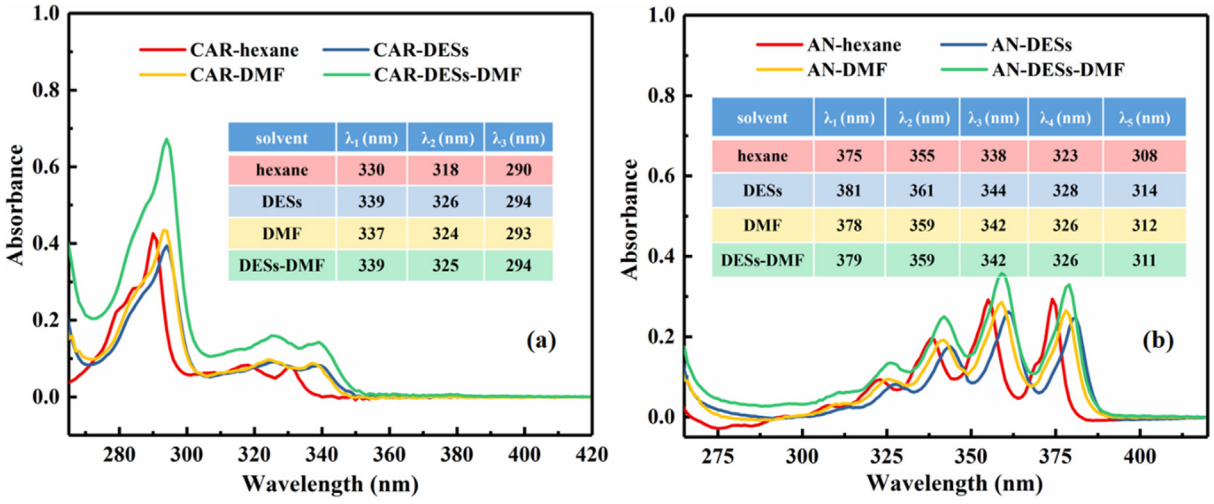


Fig. 10. UV–visible spectra for CAR (a) and AN (b) in different solvents.

between the DESs and carbazole, which are stronger than the C-H...O and C-H...π interactions present between the DESs and anthracene. The DESs component in the mixed solvent enhanced the interactions between the solvent and carbazole, but had no obvious influence on the interactions between the solvent and anthracene, thus promoting the separation of anthracene and carbazole. The impressive performance achieved by combining xylene and DESs–DMF mixed solvent system to the separation of crude anthracene oil is of major economic and environmental significance.

CRediT authorship contribution statement

Lan Yi: Methodology, Investigation, Writing - original draft. **Jie Feng:** Conceptualization, Resources. **Mario Gauthier:** Writing - review & editing. **Wen-Ying Li:** Funding acquisition, Project administration, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Authors are very grateful to National Natural Science Foundation of China (grant number: 22038008) for the financial support.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2021.116762.

References

[1] P. Liang, J. Yu, Y. Zhang, T. Jiao, X. Qin, Effects of Natural Dolomite Catalysts on Cracking Anthracene Oil, Energy Fuels 31 (11) (2017) 11765–11772, https://doi.org/10.1021/acs.energyfuels.7b02017.
[2] Z. Wang, L. Wei, X. Chen, B. Ji, X. Zhang, Excellent deep-blue emitting materials based on anthracene derivatives for non-doped organic light-emitting diodes, Opt. Mater. 58 (2016) 260–267, https://doi.org/10.1016/j.optmat.2016.05.053.
[3] Bresolí-Obach Roger, Nos Jaume, Mora Margarita, Llusa Maria, Rubén Sagristà, Anthracene-based fluorescent nanoprobes for singlet oxygen detection in biological media, Methods 109 (2016) 64–72, https://doi.org/10.1016/j.ymeth.2016.06.007.
[4] I.H. Hong, J.T. Ke, M. Toh, J. Hui, C. Kloc, Impurities in zone-refining anthracene crystals, J. Cryst. Growth 363 (1) (2013) 61–68, https://doi.org/10.1016/j.jcrysgro.2012.10.002.
[5] O., N., Pavlovich, Separation of high-purity phenanthrene and anthracene from the anthracene fraction of coal tar, Coke Chem. 55 (6) (2012) 247-248. https://doi.org/10.3103/S1068364X12060051.
[6] I. Goodarznia, F. Esmaeilzadeh, Solubility of an Anthracene, Phenanthrene, and Carbazole Mixture in Supercritical Carbon Dioxide, J. Chem. Eng. Data 47 (2) (2002) 333–338, https://doi.org/10.1021/je010093f.
[7] C.-P. Ye, X.-X. Ding, W.-Y. Li, T.-T. Wu, M.-M. Fan, J. Feng, Highly Efficient Solvent Screening for Separating Carbazole from Crude Anthracene, Energy Fuels 30 (4) (2016) 3529–3534, https://doi.org/10.1021/acs.energyfuels.5b02800.
[8] Z. Ma, X. Wei, M. Zhou, G. Liu, F. Liu, Z. Liu, X. Yu, Z. Zong, Isolation and purification of carbazole contained in anthracene slag by extraction combined with medium pressure liquid chromatography, Chin. J. Chem. Eng. 27 (12) (2019) 2925–2929, https://doi.org/10.1016/j.cjche.2018.12.012.
[9] Y.-F. Chen, Z.-M. Zong, X.-K. Li, G.-H. Liu, Z. Yang, X.-G. Jiang, F.-J. Liu, X.-Y. Wei, Q.-J. Guo, T.-S. Zhao, H.-C. Bai, B.-J. Wang, An Effective Approach for Separating Carbazole and Its Derivates from Coal-Tar-Derived Anthracene Oil Using Ionic Liquids, Energy Fuels 33 (1) (2018) 513–522, https://doi.org/10.1021/acs.energyfuels.8b02675.
[10] Y. Zheng, Industrial application and key technical problems of ionic liquids, New Chemical Materials 38 (3) (2010) 16–18.
[11] Y. Song, Research on solubilitis of anthracene and carbazole in organic solvents and a new process for crude anthracene, East China University of Science and Technology, Shanghai, 2000.
[12] C. Ye, H. Zheng, T. Wu, M. Fan, J. Feng, W. Li, Optimization of solvent crystallization process in obtaining high purity anthracene and carbazole from crude anthracene, AIChE J. 60 (1) (2014) 275–281, https://doi.org/10.1002/aic.14240.
[13] Y. Liou, C.J. Chang, Separation of anthracene from crude anthracene using gas antisolvent recrystallization, Sep. science technol. 27 (10) (1992) 1277–1289, https://doi.org/10.1080/01496399208019425.
[14] W. Li, H. Zheng, C. Ye, T. Wu, M. Fan, J. Feng, Effect of the intermolecular hydrogen bond between carbazole and N, N-dimethylformamide/isopropanolamine on the solubility of carbazole, Energy Fuels 26 (10) (2012) 6316–6322, https://doi.org/10.1021/ef301240t.
[15] M.C. Ali, Q.W. Yang, A.A. Fine, W.B. Jin, Z.G. Zhang, H.B. Xing, Q.L. Ren, Efficient removal of both basic and non-basic nitrogen compounds from fuels by deep eutectic solvents, Green Chem. 18 (1) (2016) 157–164, https://doi.org/10.1039/c5gc01823d.
[16] C. Yao, Y. Hou, S. Ren, W. Wu, Y. Ji, H. Liu, Sulfonate based zwitterions: A new class of extractants for separating phenols from oils with high efficiency via forming deep eutectic solvents, Fuel Process. Technol. 178 (2018) 206–212, https://doi.org/10.1016/j.fuproc.2018.05.031.
[17] L. Yi, J. Feng, W. Li, Z. Luo, High-Performance Separation of Phenolic Compounds from Coal-Based Liquid Oil by Deep Eutectic Solvents, ACS Sustainable Chem. Eng. 7 (8) (2019) 7777–7783, https://doi.org/10.1021/acssuschemeng.8b06734.
[18] F. Lima, M. Dave, A.J.D. Silvestre, L.C. Branco, I.M. Marrucho, Concurrent Desulfurization and Denitrogenation of Fuels Using Deep Eutectic Solvents, ACS Sustainable Chem. Eng. 7 (13) (2019) 11341–11349, https://doi.org/10.1021/acssuschemeng.9b00877.
[19] M. Larriba, M. Ayuso, P. Navarro, N. Delgado-Mellado, M. Gonzalez-Miquel, J. García, F. Rodríguez, Choline Chloride-Based Deep Eutectic Solvents in the Dearomatization of Gasolines, ACS Sustainable Chem. Eng. 6 (1) (2018) 1039–1047, https://doi.org/10.1021/acssuschemeng.7b03362.
[20] J. Chen, Y. Wang, X. Wei, P. Xu, W. Xu, R. Ni, J. Meng, Magnetic solid-phase extraction for the removal of mercury from water with ternary hydrosulphonyl-based deep eutectic solvent modified magnetic graphene oxide, Talanta 188 (2018) 454–462, https://doi.org/10.1016/j.talanta.2018.06.016.
[21] M.J. Frisch, G. Trucks, H. Schlegel, G. Scuseria, M. Robb, J. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. Petersson, Gaussian 09, Revision D. 01, Gaussian, Inc.: Wallingford, CT (2009).
[22] S.F. Boys, F.d. Bernardi, The calculation of small molecular interactions by the differences of separate total energies. Some procedures with reduced errors, Mol. Phys. 19 (4) (1970) 553-566. https://doi.org/10.1080/00268977000101561.
[23] H.L. Schmider, A.D. Becke, Chemical content of the kinetic energy density, J. Mol. Struct. (Thoechem) 527 (1–3) (2000) 51–61, https://doi.org/10.1016/S0166-1280(00)00477-2.
[24] R. Bonaccorsi, A. Pullman, E. Scrocco, J. Tomasi, The molecular electrostatic potentials for the nucleic acid bases: adenine, thymine, and cytosine, Theoretical Chemistry Accounts: Theory, Computation, and Modelling 24 (1) (1972) 51–60, https://doi.org/10.1007/BF00528310.
[25] E.R. Johnson, S. Keinan, P. Mori-Sanchez, J. Contreras-Garcia, A.J. Cohen, W. Yang, Revealing noncovalent interactions, J. Am. Chem. Soc. 132 (18) (2010) 6498–6506, https://doi.org/10.1021/ja100936w.
[26] C. Lefebvre, G. Rubez, H. Khartabil, J.C. Boisson, J. Contreras-Garcia, E. Henon, Accurately extracting the signature of intermolecular interactions present in the NCI plot of the reduced density gradient versus electron density, Phys. Chem. Chem. Phys. 19 (27) (2017) 17928–17936, https://doi.org/10.1039/c7cp02110k.
[27] R.F. Bader, Atoms in molecules, Encyclopedia comput. Chem. 1 (2002), https://doi.org/10.1021/ar00109a003.
[28] T. Lu, F. Chen, Multiwfn: a multifunctional wavefunction analyzer, J. Comput. Chem. 33 (5) (2012) 580–592, https://doi.org/10.1002/jcc.22885.
[29] W. Humphrey, A. Dalke, K. Schulten, VMD: visual molecular dynamics, J. Mol. Graph. Model. 14 (1) (1996) 33–38, https://doi.org/10.1016/0263-7855(96)00018-5.
[30] R.-H. Xiao, Coal Tar Chemistry, Metallurgical Industry Press (2002).
[31] C. Yao, Y. Hou, S. Ren, W. Wu, K. Zhang, Y. Ji, H. Liu, Efficient separation of phenol from model oils using environmentally benign quaternary ammonium-based zwitterions via forming deep eutectic solvents, Chem. Eng. J. 326 (2017) 620–626, https://doi.org/10.1016/j.cej.2017.06.007.
[32] C. Li, J. Zhang, Z. Li, J. Yin, Y. Cui, Y. Liu, G. Yang, Extraction desulfurization of fuels with 'metal ions' based deep eutectic solvents (MDESs), Green Chem. 18 (13) (2016) 3789–3795, https://doi.org/10.1039/c6gc00366d.
[33] A. Hayyan, F.S. Mjalli, I.M. AlNashef, Y.M. Al-Wahaibi, T. Al-Wahaibi, M.A. Hashim, Glucose-based deep eutectic solvents: Physical properties, J. Mol. Liq. 178 (2013) 137–141, https://doi.org/10.1016/j.molliq.2012.11.025.
[34] Q. Yang, H. Xing, B. Su, K. Yu, Z. Bao, Y. Yang, Q. Ren, Improved separation efficiency using ionic liquid–cosolvent mixtures as the extractant in liquid–liquid extraction: A multiple adjustment and synergistic effect, Chem. Eng. J. 181–182 (2012) 334-342. https://doi.org/http://dx.doi.org/10.1016/j.cej.2011.11.089.
[35] A.P. Abbott, R.C. Harris, K.S. Ryder, C. D'Agostino, L.F. Gladden, M.D. Mantle, Glycerol eutectics as sustainable solvent systems, Green Chem. 13 (1) (2011) 82–90, https://doi.org/10.1039/C0GC00395F.
[36] A. Pandey, S. Pandey, Solvatochromic probe behavior within choline chloride-based deep eutectic solvents: effect of temperature and water, Phys. Chem.

Chem. Phys. 118 (50) (2014) 14652–14661, https://doi.org/10.1021/jp510420h.
[37] A. Pandey, R. Rai, M. Pal, S. Pandey, How polar are choline chloride-based deep eutectic solvents?, Phys. Chem. Chem. Phys. 16 (4) (2014) 1559–1568, https://doi.org/10.1039/c3cp53456a.
[38] Q. Yang, S. Guo, X. Liu, Z. Zhang, Z. Bao, H. Xing, Q. Ren, Highly efficient separation of strongly hydrophilic structurally related compounds by hydrophobic ionic solutions, AIChE J. 64 (4) (2018) 1373–1382, https://doi.org/10.1002/aic.16013.
[39] P. Robinson, H. Scott, The anthracene-carbazole system, Mol. Cryst. Liq. Cryst. 5 (4) (1969) 387–404, https://doi.org/10.1080/15421406908082947.
[40] C.-P. Ye, X.-X. Ding, W.-Y. Li, H. Mu, W. Wang, J. Feng, Determination of crystalline thermodynamics and behavior of anthracene in different solvents, AIChE J. 64 (6) (2018) 2160–2167, https://doi.org/10.1002/aic.16069.
[41] S. Ding, Q. Yin, W. Du, X. Sun, B. Hou, M. Zhang, Z. Wang, Formation of Solid Solution and Ternary Phase Diagrams of Anthracene and Phenanthrene in Different Organic Solvents, J. Chem. Eng. Data 60 (5) (2015) 1401–1407, https://doi.org/10.1021/je501121v.
[42] J.F. Gonthier, S.N. Steinmann, L. Roch, A. Ruggi, N. Luisier, K. Severin, C. Corminboeuf, pi-Depletion as a criterion to predict pi-stacking ability, Chem. Commun. (Camb.) 48 (74) (2012) 9239-41. https://doi.org/10.1039/c2cc33886f.
[43] P. Politzer, J.S. Murray, The fundamental nature and role of the electrostatic potential in atoms and molecules, Theor. Chem. Acc. 108 (3) (2002) 134–142, https://doi.org/10.1007/s00214-002-0363-9.
[44] R.F.W. Bader, Atoms in molecules, Acc. Chem. Res. 18 (1) (1985) 9–15, https://doi.org/10.1021/ar00109a003.
[45] P.L.A. Popelier, Characterization of a Dihydrogen Bond on the Basis of the Electron Density, J. Phys. Chem. A 102 (10) (1998) 1873–1878, https://doi.org/10.1021/jp9805048.